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ML; DL, NLP
Boot camp

Krushnak Sir

sem-20

lec-3 Population vs sample

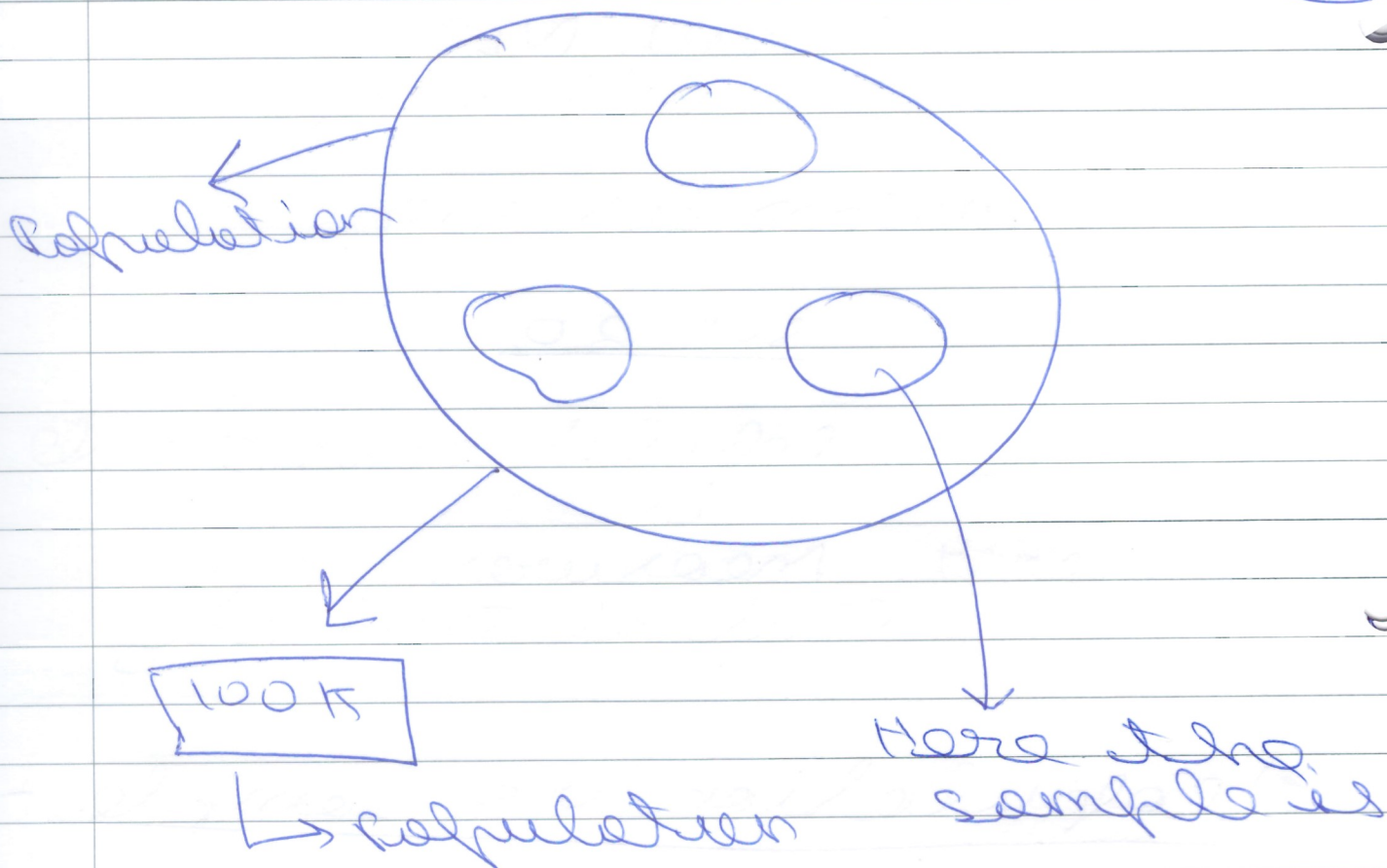
lec-4 measures of
central tendency

⇒ Population vs sample:-

Suppose an amount of
population is given
here.

→ And we have to
count all the
weights of all
people.

Now it's important
to calculate the
weights of all people.



Here the sample of the population is taken

and then the weights are counted using the sample.

the population is represented as N

the sample is represented as n .

- For example the same criterion is followed by all Elected Members.

Measures of Central Tendency

- Three Measures of Central Tendency are

- (1) Mean
- (2) Median
- (3) Mode

Mean is basically the average

$$\Rightarrow \frac{\text{Total sum}}{\text{Total Number}}$$

Mean Formulae

N (Population Data)

$$\mu = \sum_{i=1}^N \frac{x_i}{N}$$

Mean for Population

n (Sample Data)

$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n}$$

Mean for Sample

Let's take an example

Ages = 2, 1, 3, 4, 5 → Distribution

$$\mu = \frac{1+3+4+5}{4}$$

$$= \frac{13}{4} = 3.25$$

↓
This is the average of the particular distribution

$$\mu = \frac{1+3+4+5}{4}$$

→ If we're told in a question that this

(5)

⇒ New set is still about
median:

For that set is made
a minor change in
the set.

In the data, as we
can see that.

↪ Age = { 1, 3, 4, 5, 100 }

↘ This 100
outlier

Now if mean is calculated

$$\bar{x} = \frac{1+3+4+5+100}{5}$$

$$= \frac{113}{5} = 22.6$$

↪ Here the mean
has created a lot of
difference as we can see

Now if we calculate
the value, then it
will be

Suppose the data given to us is:-

Ages = 2, 4, 3, 1, 5, 10, 0, 6

Sort the numbers
 in increasing order
 $\Rightarrow \{ 1, 3, 4, 5, 100 \}$
 add no.
 of numbers.

Now median is calculated as:-

Median $\rightarrow 4$

Here mention is quite near to the mean.

- Now suppose if an
no. of numbers
numbers = $\{1, 3, 4, 9, 100,$
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p. 711

(2)

Now if we calculate the median, it comes out to be:-

Out of 2 numbers of 2 middle values

$$\Rightarrow \frac{4+5}{2} = \frac{9}{2} = 4.5$$

This is also equal to 4.5
the ~~average~~ average

(3) Mode:-

Here data is given as $\{4, 3, 2, 1, 1, 4, 4, 5, 2\}$
100%

In mode we select the element which has the maximum frequency.

The result = 4

Here also 100 is not a value which can be used if mean is calculated.